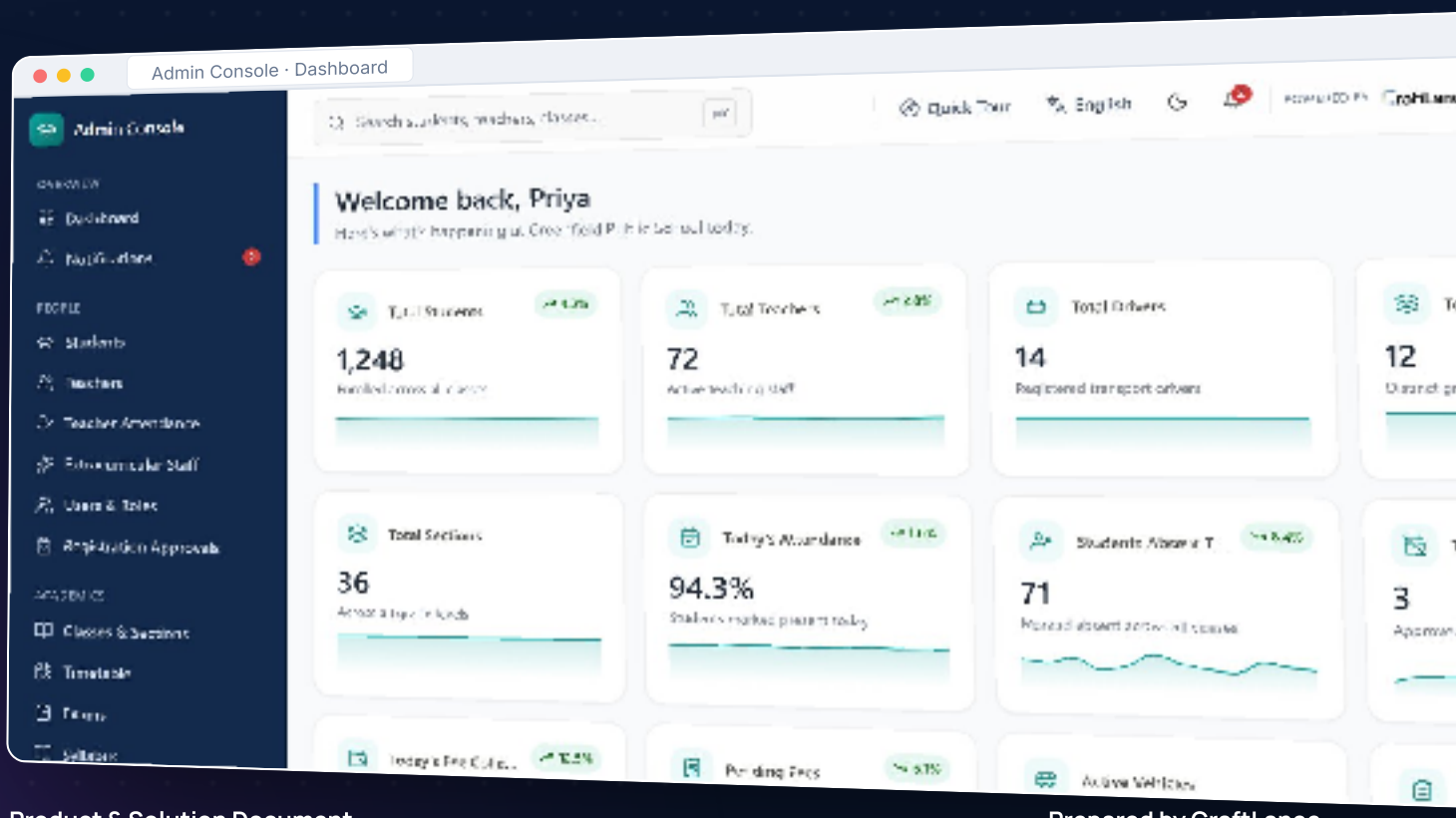
Smart School Management
SCHOOL ERP PLATFORM

CraftLanee

Every school. Every role. One platform.

A complete, cloud-hosted school management system — academics, attendance, fees, live GPS transport, communication and student engagement — delivered through purpose-built portals for everyone who runs, teaches at, or studies in your school.

Multi-school ready | 9 role-based portals | Live bus tracking | Web · Tablet · Mobile | 5 languages



Product & Solution Document
Version 1.0 · September 2026

Prepared by CraftLanee
For prospective school partners

INSIDE THIS DOCUMENT

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This document explains what the Smart School Management System does, who it serves, how it looks and works in practice, and how it is built and delivered. Every screen shown is the real product, captured with sample data for a demonstration school.

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About the screenshots. All screens in this document were captured from the live application running against a fictional demonstration school, "Greenfield Public School". Names, figures and locations are sample data used for illustration only.

01 AT A GLANCE

One system of record for the whole school

The Smart School Management System replaces registers, spreadsheets, phone calls and scattered messaging groups with a single, secure web platform. Administrators, principals, teachers, students and parents, accountants, drivers and activity instructors each sign in to a portal designed around their own daily work — and every portal draws on the same, always-current school data.

9
Role-based portals, each with its own dashboard

20+
Integrated modules, from admissions to transport

400+
Secure API operations powering the platform

5
Interface languages, plus light and dark themes

What makes it different

Live bus tracking, built in

Parents see the school van move on a map with a live ETA, and get automatic alerts at 10 minutes, 5 minutes and on arrival. Drivers need nothing more than a phone browser.

Run many schools from one place

A platform tier lets a school group create, activate and oversee several campuses, each with fully isolated data, from a single login.

Approvals instead of loose ends

Homework, leave, registrations and profile changes each move through a defined review chain, with a record of who approved what and when.

Students actually enjoy using it

A friendly, animated student portal with 17 learning games, badges and streaks sits alongside homework, marks, timetable and fees.

Available everywhere your people are

Runs in any modern browser on desktop, tablet or phone — nothing to install. The interface can be switched between **English, Hindi, Telugu, Kannada and Tamil**, supports **light and dark themes**, and includes a built-in **Quick Tour** that walks each new user through their own menu.

02 WHY SCHOOLS NEED IT

From scattered tools to one source of truth

Most schools run on a patchwork: paper registers, fee notebooks, spreadsheets, WhatsApp groups and phone calls to the bus driver. Each works on its own; together they leave gaps that cost staff time and parents' trust.

Area	The usual way	With Smart School Management
Attendance	Paper registers, reconciled by hand at month-end	Marked in seconds per class; daily, monthly and student-wise reports are instant
Fees	Ledgers and spreadsheets; no live view of dues	Class-wise and per-student fee plans, instant receipts, collection dashboards by day, week, month and year
School transport	Parents call the driver; no record of who boarded	Live map with ETA, automatic proximity alerts, and pickup/drop marking for every child
Homework & results	Relayed verbally or via a notebook that may not reach home	Published to the student portal with notifications; marks released only when teachers publish them
Leave requests	Phone calls and verbal approvals, no audit trail	Online requests routed to the right approver, with cut-off rules and a full history
Announcements	Printed circulars and messaging groups	Targeted by class, role or person; scheduled publishing; delivered in-app, by email and push
New accounts	Paper forms, re-typed by office staff	Self-registration with approval by the right reviewer before any access is granted
Multiple campuses	Separate systems, or none; no group-level view	One platform, isolated data per school, consolidated oversight for management

The result: information is entered once and is immediately visible to everyone who is authorised to see it — and to no one who isn't.

03 ONE PLATFORM, NINE PORTALS

A dedicated experience for every role

Each person signs in and lands on a dashboard built for their job. Menus, data and actions are tailored to the role — and enforced on the server, so no one can see or change more than they should.

Super Admin
Runs the platform: creates and activates schools, assigns school administrators, reviews new-school requests, and sees group-wide statistics and a full audit log.

School Admin
Full control of a school: students, staff, classes, timetable, exams, fees, transport, announcements, users, reports and settings.

Principal
The same powerful admin console, locked to their own school. Approves staff registrations and staff leave; applies for their own leave.

Teacher
Attendance, homework, marks, question papers, syllabus and timetable for assigned classes. Class teachers also review homework, student leave and profile changes.

Student & Parent
One shared family login: attendance, homework, timetable, marks, report cards, fees, live bus tracking, activities, learning games and leave requests.

Accountant
Fee structures, payments, receipts, per-student adjustments and collection reports — with no access to academic records.

Driver
A mobile-first screen to start a trip, share live GPS, and mark each child picked up, absent or dropped.

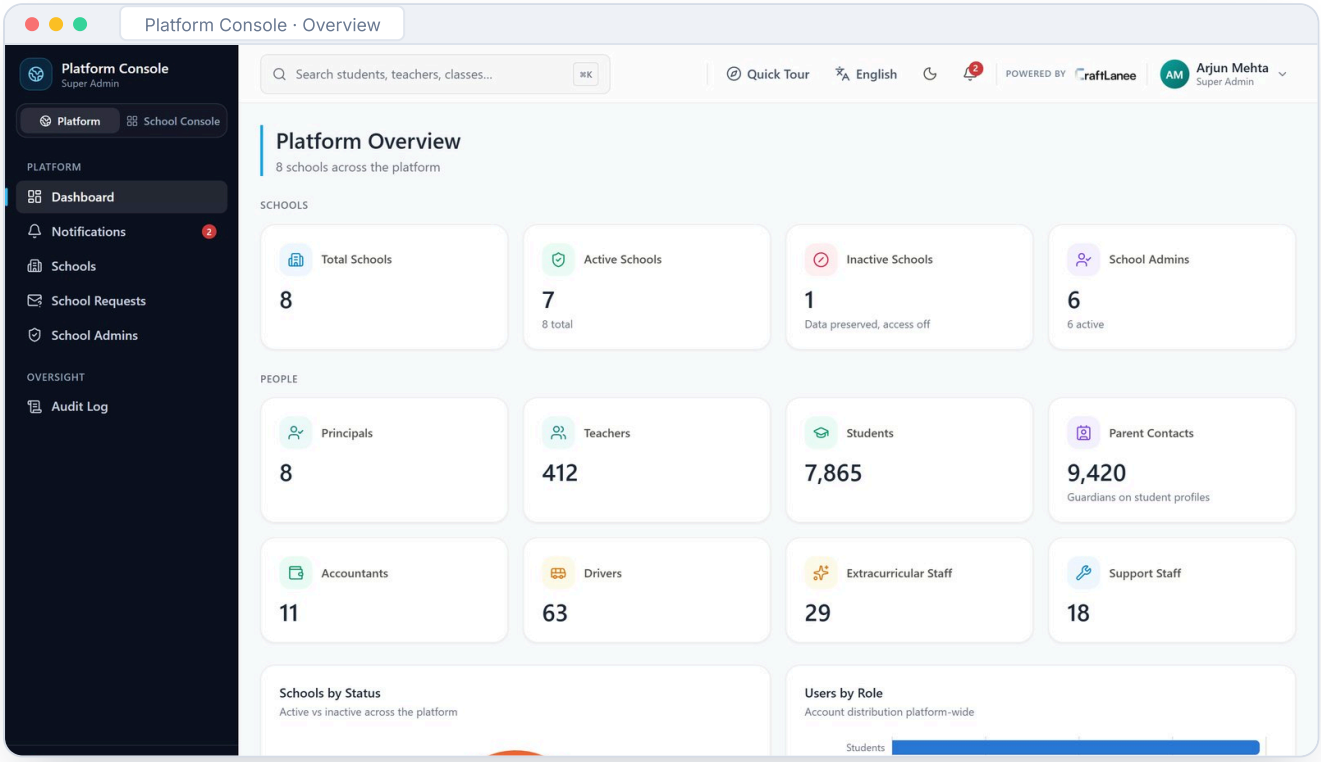
Extracurricular Staff
Instructors for dance, yoga, karate, music and more manage batches, schedules, session attendance, practice work, events and achievements.

A ninth **Support Staff** role is also available for non-teaching staff who need a self-service profile. One person can hold more than one role where needed.

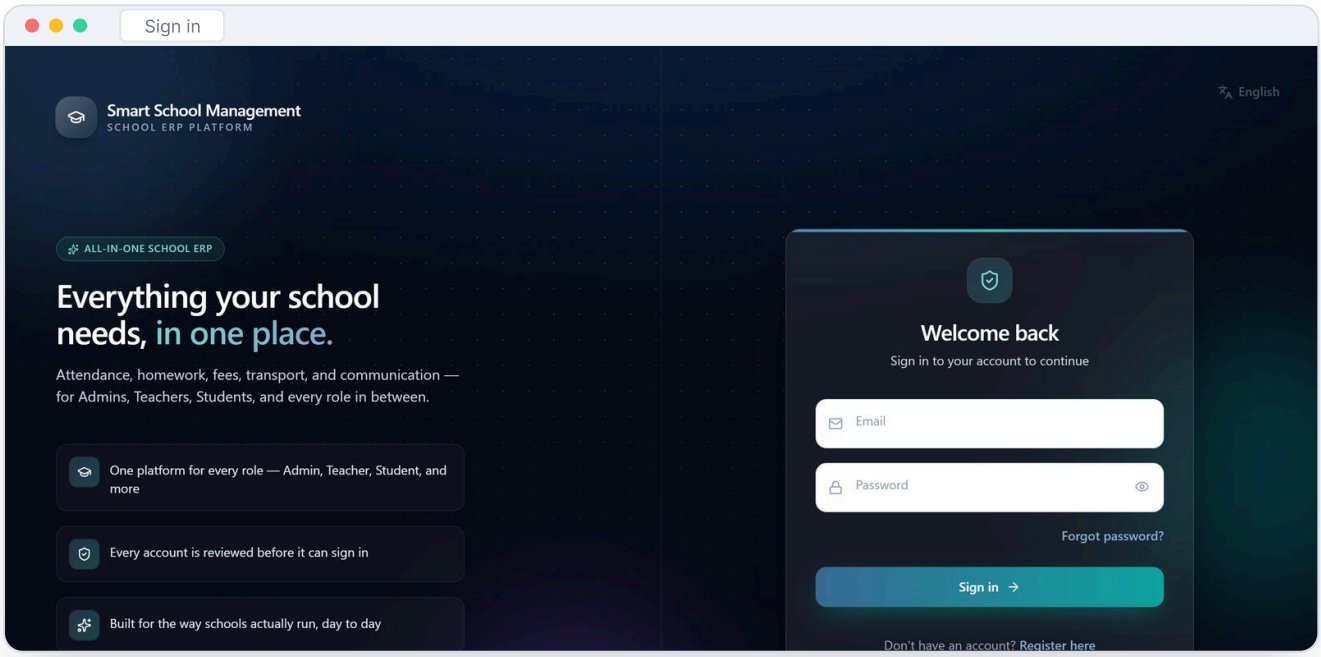
04 PRODUCT TOUR

Leadership: the whole picture at a glance

Group management and school leadership get live dashboards — no waiting for month-end reports.



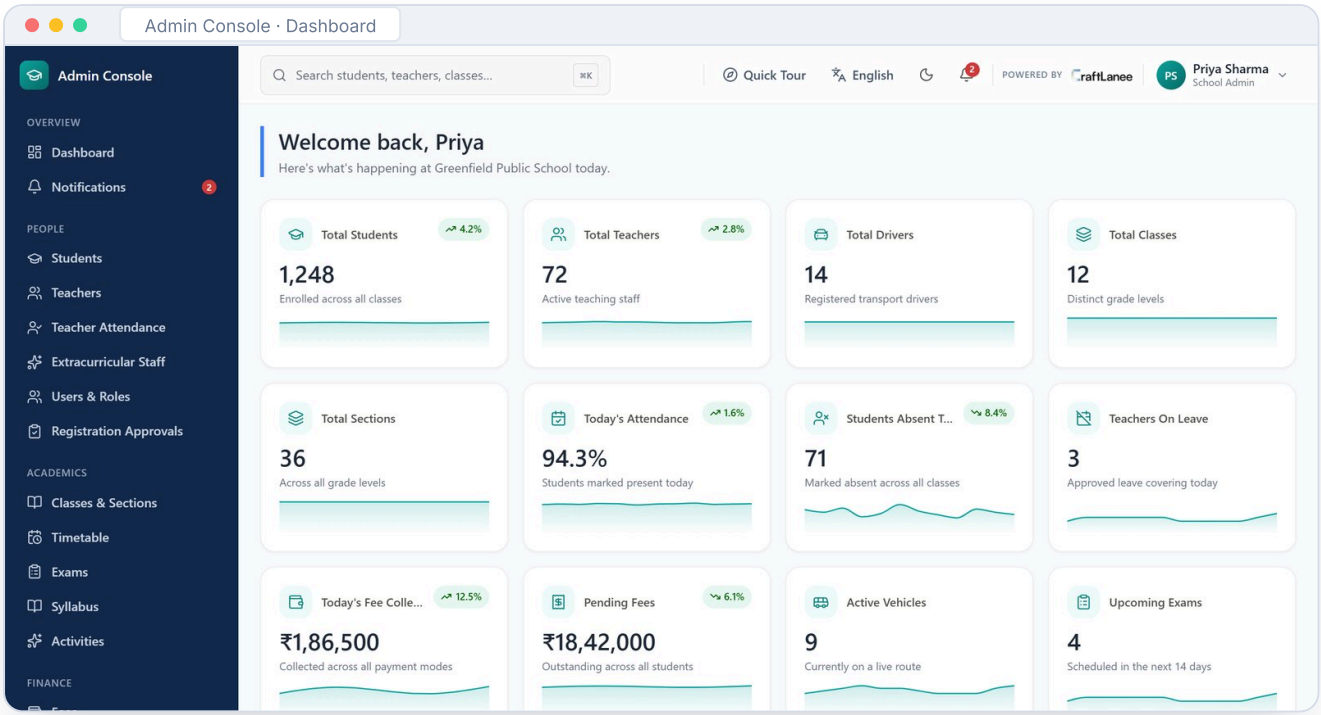
Platform Console Super Admins see every school in the group — active and inactive schools, administrators, and head-counts by role — with charts of users per role and school status.



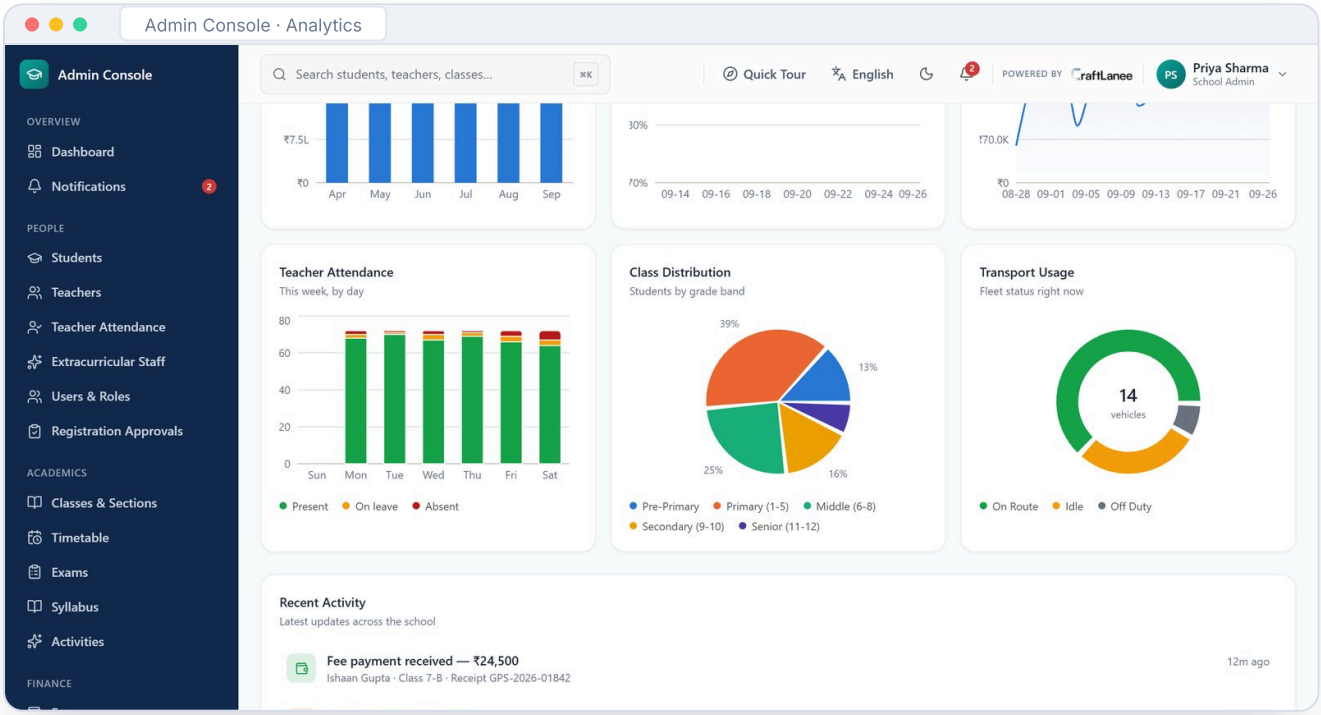
Secure sign-in A single sign-in page for every role. New users can register themselves; their account stays locked until the right person approves it.

School Admin & Principal — Admin Console

The console opens on twelve live indicators — enrolment, staff, today's attendance, absences, staff on leave, today's fee collection, pending dues, vehicles on route and upcoming exams — each with a trend line.



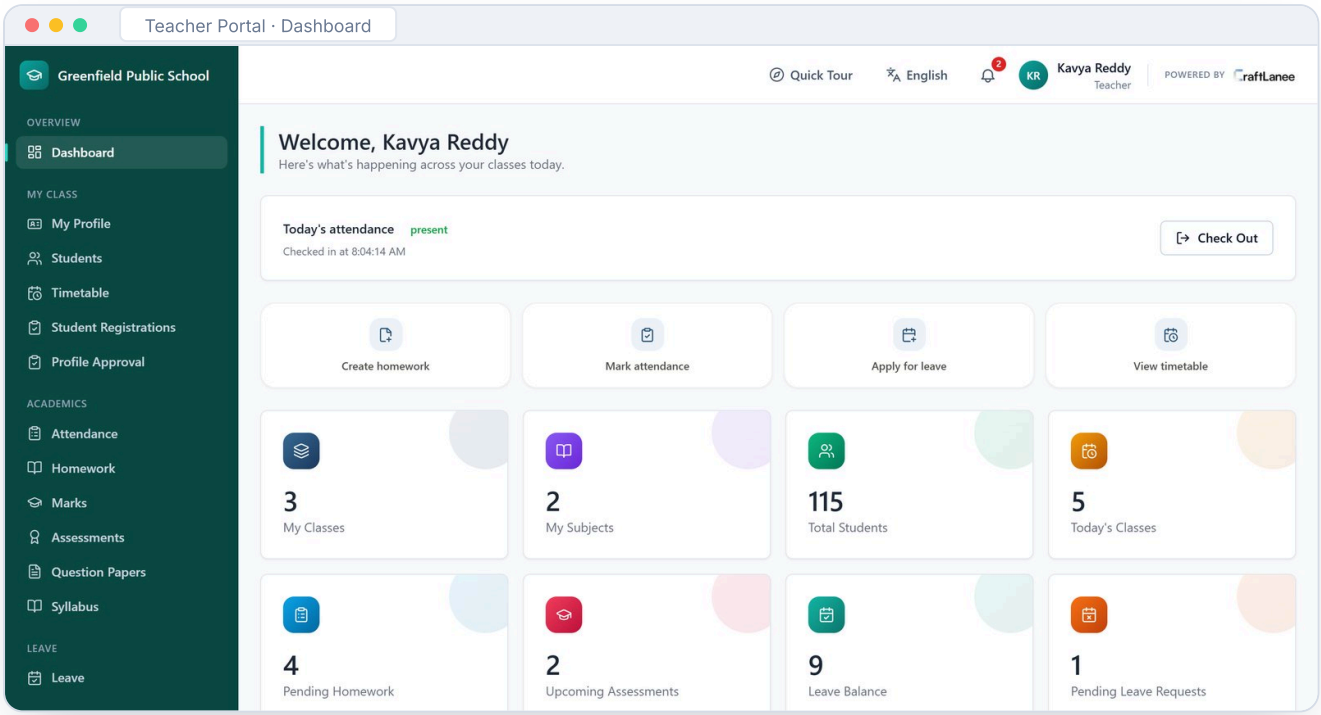
Dashboard Key numbers with trend indicators. Every card links through to the full module behind it.



Analytics Fee collection by month, daily attendance, weekly teacher attendance, class distribution, fleet status and a live feed of recent activity across the school.

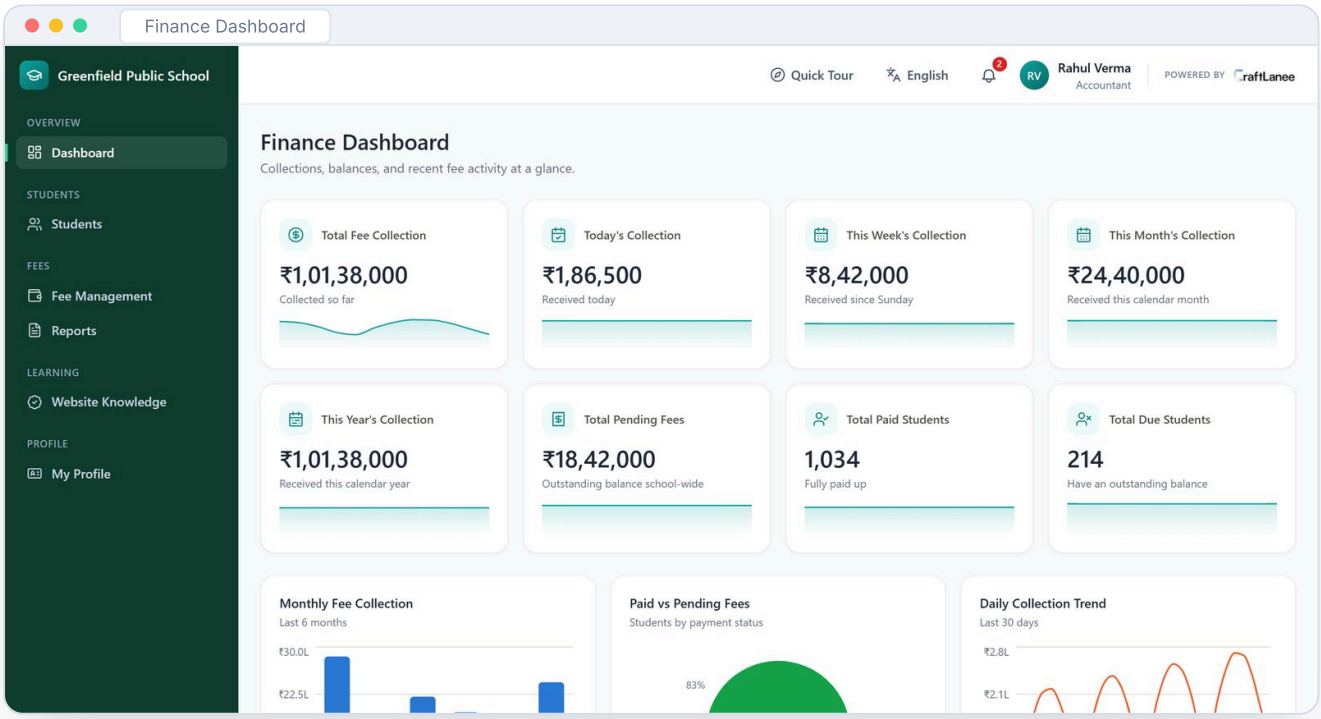
Teachers — everything for their classes, nothing extra

Teachers check in for the day, see their classes and subjects, and jump straight to attendance, homework, marks and leave. Class teachers get additional review queues for their own class.



Teacher dashboard Self check-in/check-out, quick actions, and counts for classes, students, today's periods, pending homework, upcoming assessments and leave balance.

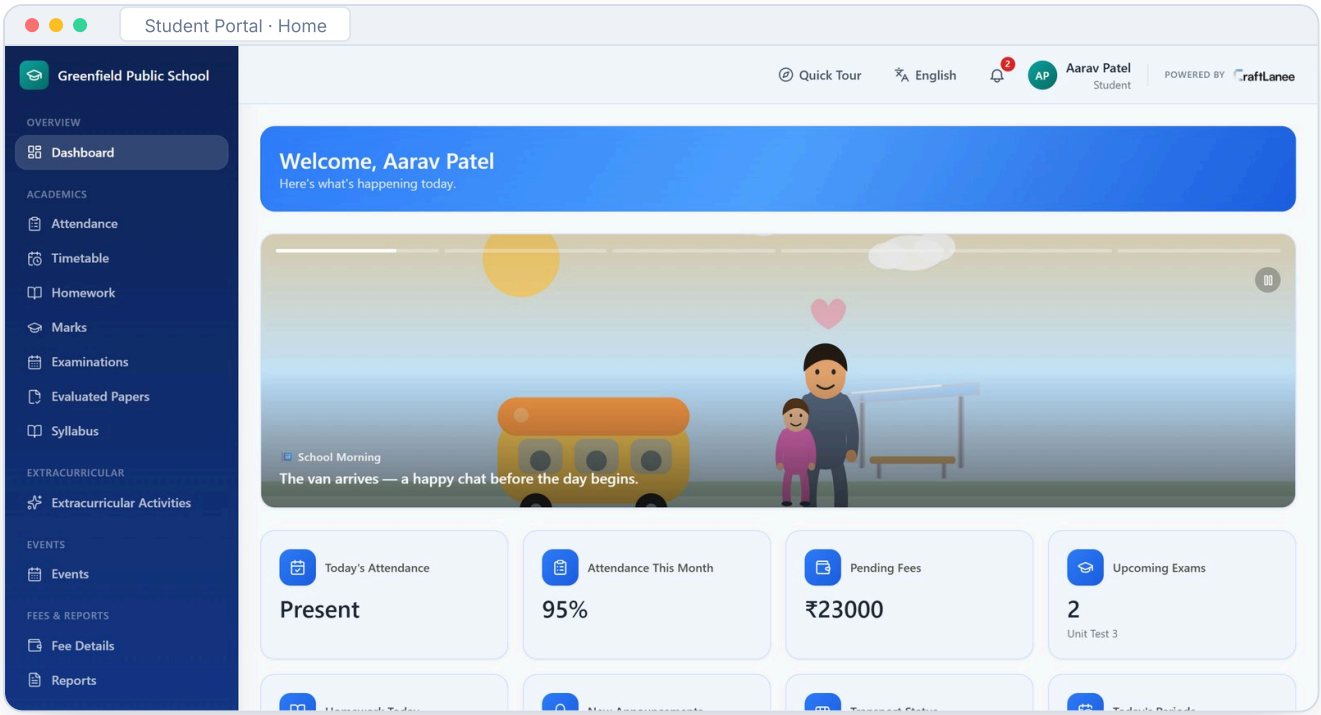
Accountants — a finance desk that balances itself



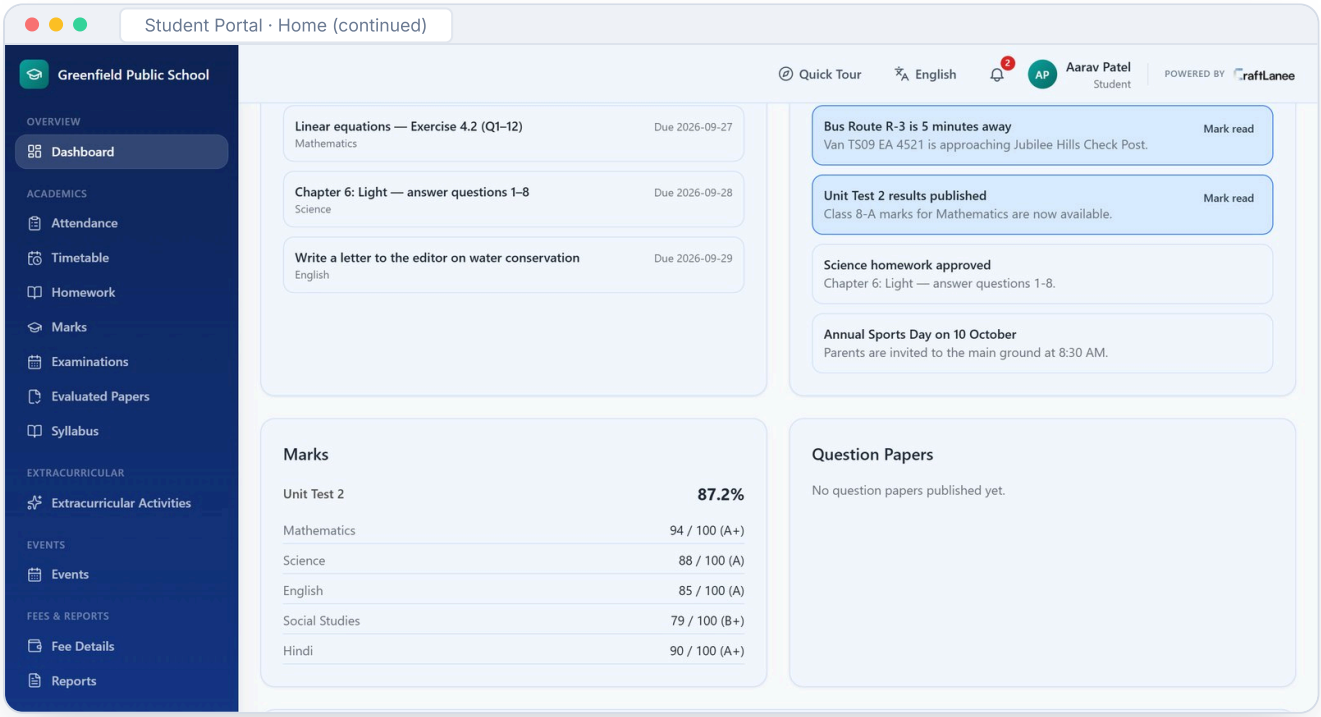
Finance dashboard Collections for today, this week, month and year; total pending; paid vs. due students; monthly collection, paid-vs-pending split and a daily collection trend.

Students & parents — the family portal

A warm, friendly portal — distinct from the staff consoles — with an animated story banner and everything a family checks each day: attendance, homework, dues, exams, announcements, the bus and today's periods.



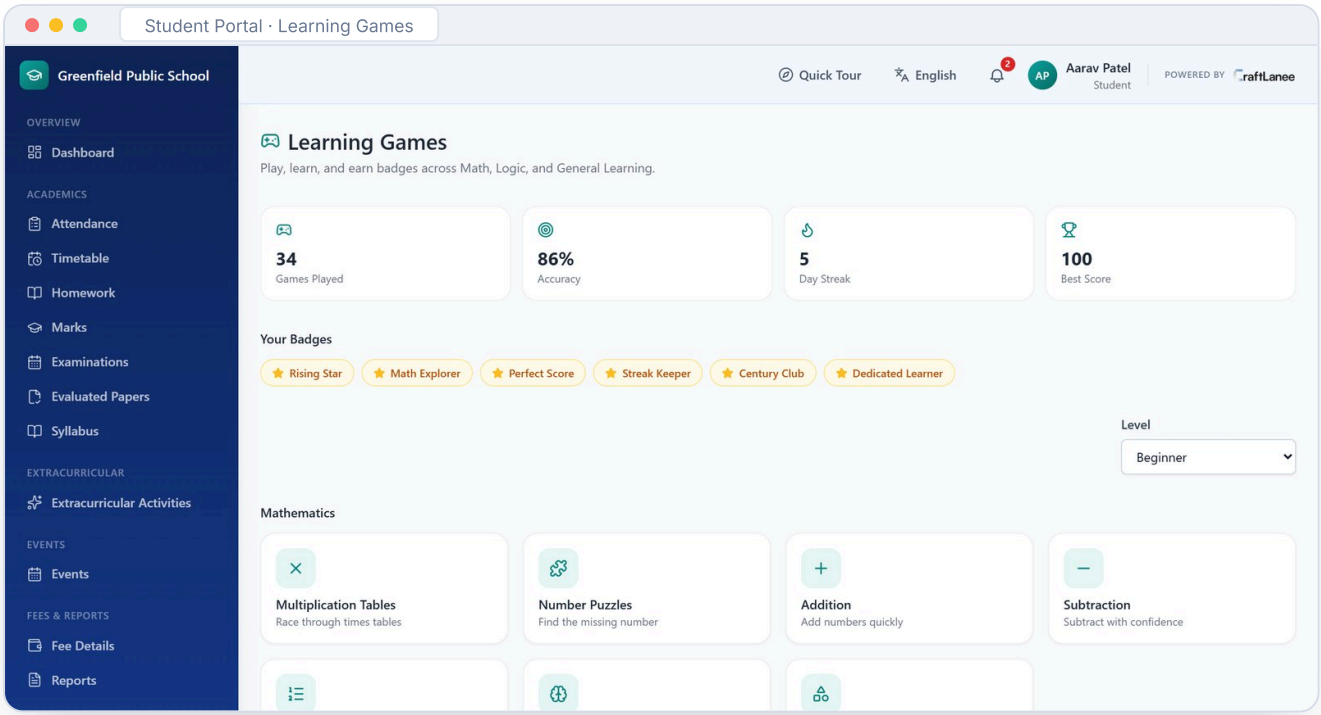
Portal home Today's attendance, monthly attendance %, pending fees, upcoming exams, homework, announcements, transport and the day's periods.



Details on one screen Homework with due dates, unread announcements, latest exam marks with subject grades, and today's timetable.

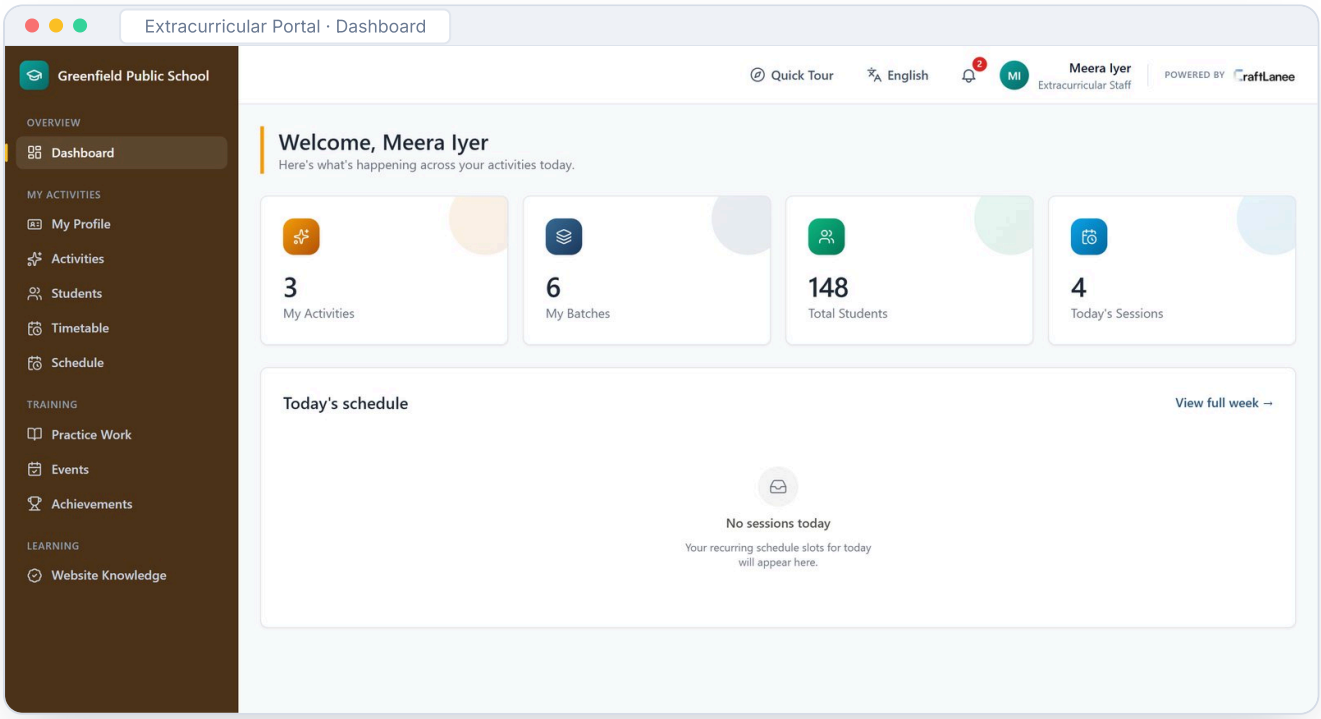
Learning Games — practice that feels like play

Students build skills through 17 games in Mathematics, Logic and General Learning, at three difficulty levels, earning badges and keeping daily streaks.



Learning Games Games played, accuracy, day streak and best score, earned badges, and the game library by category and level.

Extracurricular staff — activities, organised

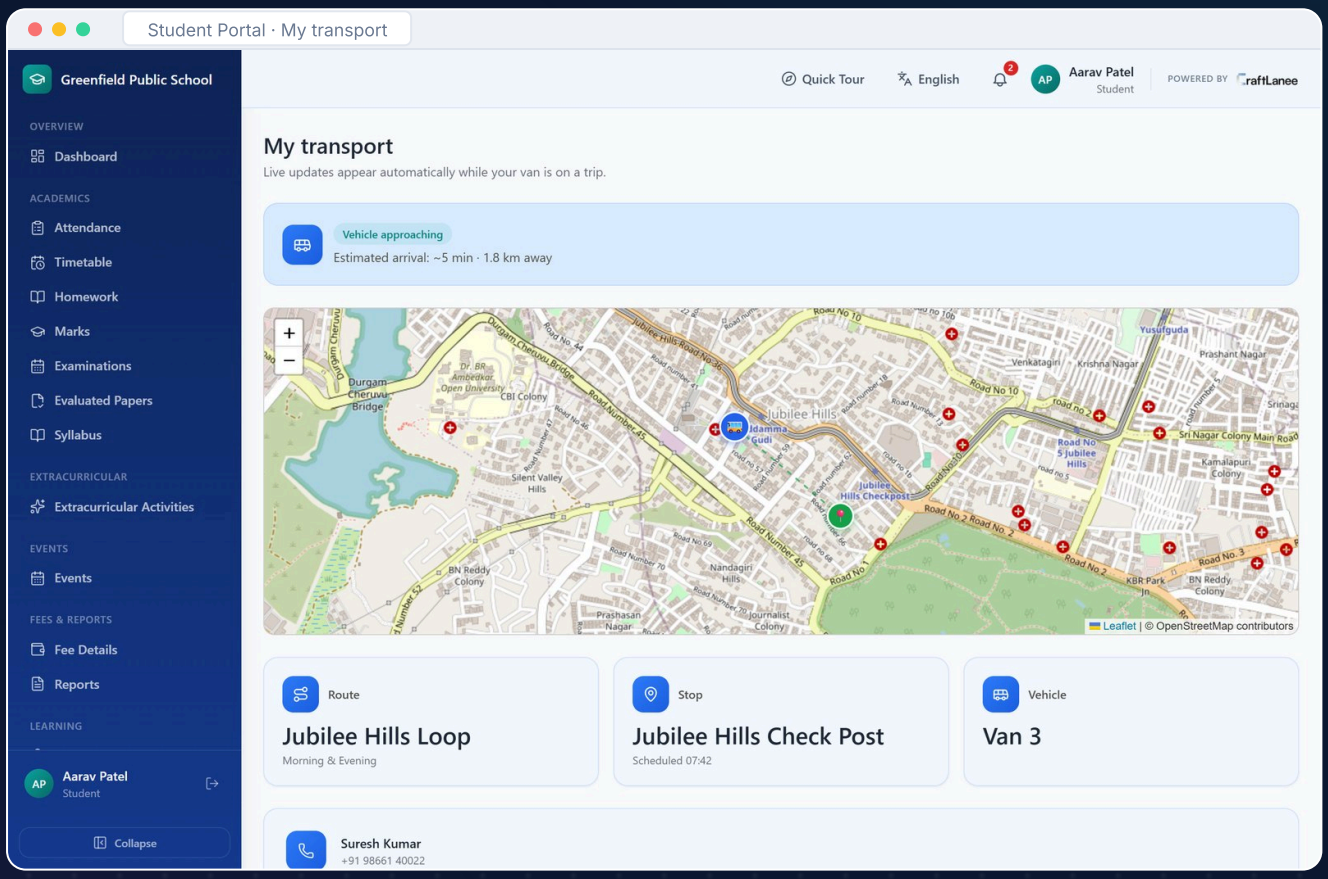


Instructor dashboard Activities, batches, students and today's sessions, with menus for schedule, attendance, practice work, events and achievements.

05 SIGNATURE FEATURE

Live GPS transport.
No more waiting at the gate.

The school van becomes visible to every family on its route. Drivers share their location from an ordinary phone browser; parents watch the van approach on a live map, see a real ETA, and are alerted automatically as it gets close. The school sees its whole fleet at once.



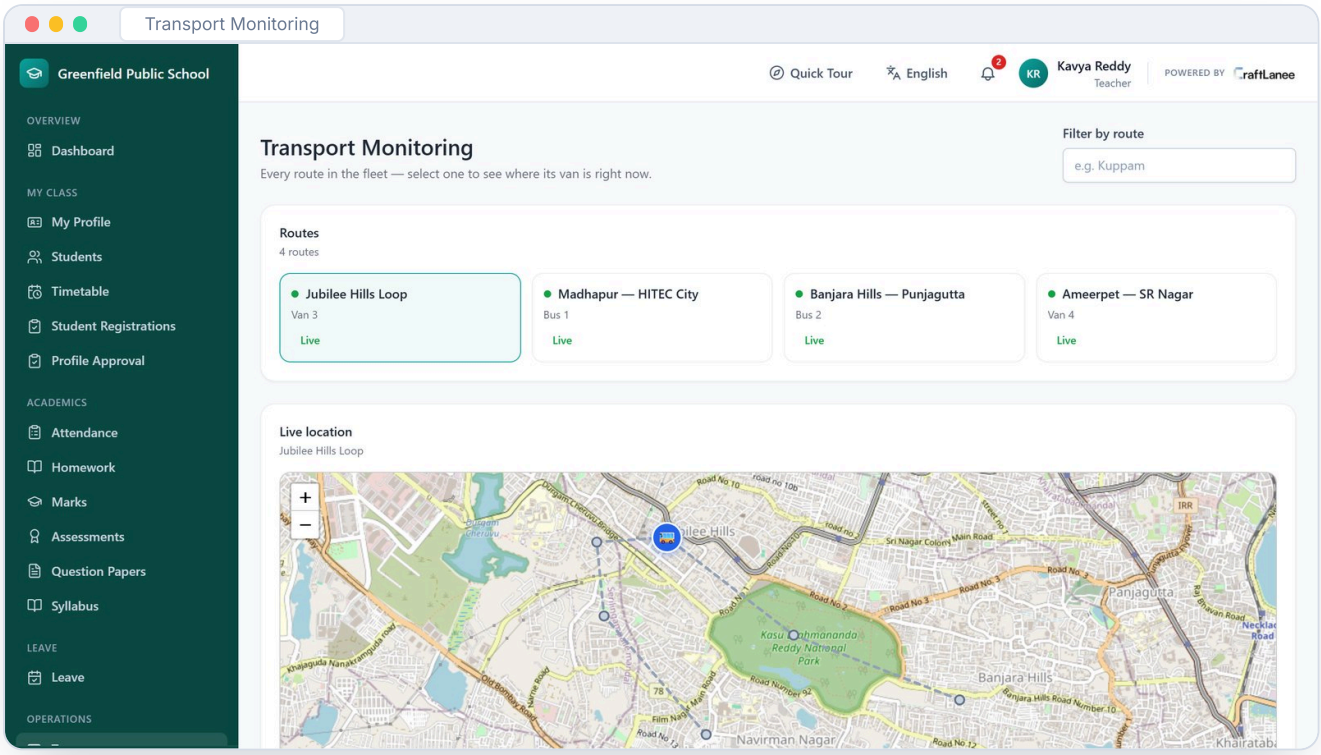
What parents see Trip status ("Vehicle approaching"), ETA and distance, the van's live position and the family's own position, route, stop, vehicle and a one-tap driver contact card.

10 min
First automatic alert as the van approaches the child's stop

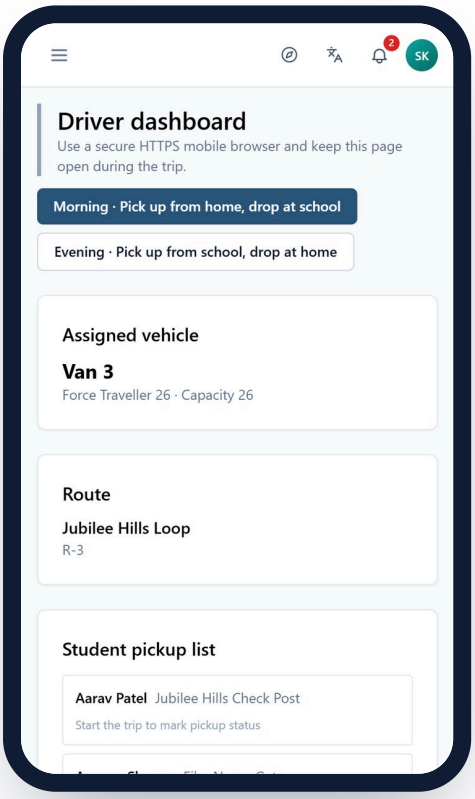
5 min
Second alert — time to head out to the stop

Arrived
Final alert when the van reaches the stop — each sent once, never spammed

The school's view — the whole fleet, live



Transport Monitoring Every route with a live indicator. Selecting a route shows the van's position against its ordered stops, plus picked-up / absent / remaining counts for the trip.



The driver's view — one phone, one tap

Drivers use a simple, mobile-first screen. No app store, no special device.

- **Morning and evening trips** — the evening route automatically runs the morning stops in reverse.
- **Start / end trip** with the screen kept awake, so tracking never silently stops.
- **Location every ~10 seconds**, with an offline queue that catches up when signal returns.
- **Pickup list by stop** — mark each child picked up or absent, and dropped on the way home.
- **Trip recovery** — reopening the page resumes an in-progress trip.

> Fleet records are included too: vehicles, drivers and licences, routes and stops, student-to-stop assignment, transport fees and a vehicle maintenance log.

How a trip works

Every step updates the family portal and the school's monitoring screen at the same time.

1 · Trip starts
Driver taps Start. Families on the route are notified.

›

2 · Live GPS
Position shared every ~10 s and shown on the map.

›

3 · ETA & alerts
Distance and ETA per stop; 10-min, 5-min and arrival alerts.

›

4 · Pickup
Driver marks each child picked up or absent.

›

5 · Arrival
Trip ends at school; history is saved.

What the family sees, stage by stage

Waiting for pickup · Vehicle approaching · Picked up · Reached school · Return trip started · Returning home · Dropped at home · Absent

Capability	How it works
ETA calculation	Distance to each stop is calculated from live GPS, with speed averaged over the van's recent readings for a realistic arrival estimate.
No duplicate alerts	Each alert (10 min, 5 min, arrived) is recorded per child per trip, so frequent GPS updates never trigger repeat messages.
Privacy by design	Families see only their own child's van, and only while a trip is running. Access closes the moment the driver ends the trip.
Trip history	Every trip, with pickup and drop status per child, is kept for review by the school and visible to the family.
Hardware	No GPS boxes to buy. Any smartphone with a modern browser works as the tracker; maps use OpenStreetMap.

Fleet management, included

Vehicles
Registration, capacity, insurance and fuel type for every bus and van.

Drivers
Profiles, licence numbers and expiry dates; a primary and backup driver per route.

Routes & stops
Ordered stops with pickup times; each student assigned to a stop, morning, evening or both.

Fees & upkeep
Transport fees tracked per student, and a maintenance log with next-service reminders.

06 MODULE CATALOGUE

Everything included

All modules share one database and one login. There are no separate add-ons to integrate.

Students

- Complete profiles: personal, parent, medical, Aadhaar, address
- Documents, siblings, photos
- Bulk import, activate / deactivate

Teachers & staff

- Profiles, qualifications, employee IDs
- Class and subject assignments; class teachers
- HR documents, daily check-in and absence tracking

Classes & academics

- Academic years, branches, departments
- Classes, sections, capacity, subjects
- Holidays and school calendar

Attendance

- Fast class-wise marking with edit history
- Daily, monthly, student-wise, class-wise reports
- Attendance % and calendar for families

Timetable

- Weekly period builder with rooms
- Academic and activity periods
- Teacher change requests with approval

Homework

- Assignments with attachments and due dates
- Class-teacher review before publishing
- Student submissions and tracking

Exams & marks

- Exams for one, several or all classes
- Exam timetable, question papers, marks entry
- Auto grades, publishing, report cards (PDF)

Evaluated papers

- Upload graded answer sheets per student
- Families view them per exam and subject

Syllabus

- Per class, subject and academic year
- Publish / unpublish; secure preview and download

Fees & finance

- Class-wide or per-student fee plans
- Discounts, scholarships, overrides, transport fees
- Payments, receipts, collection reports

Transport

- Vehicles, drivers, routes, stops
- Live GPS tracking and alerts
- Trip history and maintenance log

Announcements

- Rich-text notices with attachments
- Target by class, role or specific staff
- Scheduled publishing; emergency alerts

Leave management

- Student, teacher and principal leave
- Right approver for each; 6 PM same-day cut-off
- Entitlements per school (casual, sick, other)

Registration & approvals

- Self-registration for six roles
- Accounts stay locked until approved
- Status page for applicants

Users & roles

- Create, bulk-manage and deactivate users
- Role assignment; multiple roles per person

Reports hub

- Attendance, students, teachers, fees, transport, exams
- Export to PDF and Excel

Extracurricular

- 16 ready-made activities, plus your own
- Batches, schedules, attendance, events
- Achievements and certificates gallery

Learning Games

- 17 games across Math, Logic, General
- 3 levels, 10 badges, daily streaks

Website Knowledge certification

- Role-specific quizzes on using the platform
- Pass mark, retakes, printable certificate
- Progress monitoring by supervisors

Notifications

- In-app, email and browser push
- 30+ event types, delivered automatically

Multi-school platform

- Create, activate, suspend schools
- Admins across several schools; school switcher
- Group statistics and audit log

Settings

- School profile and logo, calendar
- Academic structure, notification preferences

Audit & security logs

- Login history, admin activity log
- Platform-level audit trail

Usability

- 5 languages, light / dark mode
- Quick Tour onboarding, global search

Extracurricular activities included out of the box: Dance, Yoga, PT, Karate, Music, Singing, Violin, Keyboard, Guitar, Drawing, Painting, Chess, Skating, Drama, Spoken English and Martial Arts — each school can add its own.

07 BUILT-IN APPROVAL WORKFLOWS

Accountability, without the paperwork

Sensitive changes never go straight live. Each one is routed to the right reviewer, and every decision is recorded and notified back to the person who asked.

Homework

Subject teacher
Creates homework for a class

›

Class teacher
Approves, or requests changes with a note

›

Students & parents
See it in the portal and are notified

Homework created by the class teacher is published immediately.

Leave

Student leave
→ their own class teacher

•

Teacher leave
→ the principal

•

Principal leave
→ school admin (never another principal)

Same-day requests close at 6 PM; duplicate requests are blocked; two approvers clicking at once can never double-process a request.

New account registration

Student
→ class teacher of the chosen class

•

Teacher, accountant, driver, instructor
→ principal

•

Principal
→ school admin

Until approved, the account cannot sign in. Class and subject assignments requested by a new teacher take effect only on approval.

Other reviewed changes

Student profile edits
Proposed by the family, approved by the class teacher before they change the record.

Timetable changes
Teachers propose changes to their periods; administrators approve them.

New schools
School admins request an additional campus; the Super Admin approves and creates it.

08 COMMUNICATION & NOTIFICATIONS

The right message, to the right person, automatically

One notification engine drives three channels. Staff don't have to remember to tell anyone — the system does it as part of each action.

In-app

A live notification bell and full history in every portal, with read tracking and one-click navigation to the item.

Email

Sent automatically alongside in-app notices; each school can switch email on or off.

Browser push

Instant pop-up alerts on phone or desktop for high-priority events like leave decisions, results and fees.

Event	Who is notified
Homework published or approved	Students and families of the class
Marks, results, question papers published	Students of the class; the class teacher sees subject-wise completion and can remind colleagues
Fee due, updated, or payment recorded	The student's family, plus class teacher, principal and administrators
Bus trip started, 10 / 5 min away, arrived, picked up, dropped	Families of students on that route
Leave submitted / decided	The designated approver / the person who applied
Registration submitted / decided	The assigned reviewer / the applicant
Announcements	Whole school, specific classes, roles or named staff — immediately or at a scheduled time
Emergency alert	Everyone in the school, marked as critical

09 LEARNING & ENGAGEMENT

More than administration

The platform gives students reasons to log in every day, and helps every staff member become confident with the system.

Learning Games

Mathematics · 7

Multiplication Tables, Number Puzzles, Addition, Subtraction, Number Sequences, Mental Math, Pattern Recognition

Logic · 5

Matching Puzzles, Memory Cards, Missing Numbers, Logical Sequences, Shape Matching

General Learning · 5

Word Matching, Vocabulary, Spelling, Memory Activities, General Knowledge

Beginner, Intermediate and Advanced levels · accuracy and best-score tracking · daily streaks · 10 badges including *Rising Star, Math Explorer, Puzzle Master, Perfect Score, Century Club* and *Grand Master*.

Website Knowledge certification

- Role-specific multiple-choice quizzes about using the platform — for students, teachers, principals, admins, accountants, drivers and instructors
- Questions and options shuffled on every attempt; retakes rotate through different question sets
- Configurable pass mark (70–90%)
- A printable certificate on passing, with certificate number and score
- Supervisors track completion: class teacher → students, principal → teachers and staff, admin → principals
- Question bank managed centrally by the Super Admin

Extracurricular activities

- Batches by activity, class and year — whole class or selected students
- Weekly schedules with clash checking for instructors
- Session attendance, separate from academic attendance
- Practice work and events for each batch
- Achievements and certificate uploads per student
- A photo gallery of achievements in the family portal

10 SECURITY & DATA PRIVACY

Your school's data stays your school's

Student records are among the most sensitive data a school holds. Protection is built into every layer — not added afterwards.

Secure sign-in

Industry-standard token-based authentication. Passwords are never stored by the application; they are handled by a managed, hashed identity service. Login attempts are rate-limited and logged.

Role-based access control

43 granular permissions across the roles, checked on the server for every request — plus ownership checks, so a teacher can only act on their own classes and a family only on their own child.

Complete isolation between schools

Every record belongs to a school. Database-level row security, in addition to server checks, keeps one school's data invisible to every other school on the platform.

Private documents

Student and staff documents, exam papers, evaluated scripts and syllabus files are stored privately and served only through short-lived, expiring links.

Full audit trail

Login history, school administrative activity and platform-level actions are all recorded for review.

Safe deactivation

Accounts are deactivated without losing history. Suspending a school suspends access for its users instantly, after an impact preview.

All traffic is encrypted over HTTPS. Every form is validated on the server, API endpoints use hardened security headers and strict cross-origin rules, and data is held in a managed PostgreSQL database with automatic daily backups.

11 TECHNOLOGY & ARCHITECTURE

Modern, cloud-native, built to scale

A proven, widely supported technology stack — fast for users, straightforward to host, and ready to grow from one school to many.

Users
Any browser, any device

Desktop · Tablet · Smartphone · Driver phone GPS · Browser push

▼ HTTPS

Web application
React 19 · TypeScript

Role-gated portals · Tailwind CSS design system · Leaflet live maps · Recharts analytics · PDF & Excel export · 5 languages

▼ Secure REST API (/api/v1)

API server
Node.js · Express · TypeScript

36 modules · 400+ operations · Permission checks on every call · Input validation · Rate limiting · Schedulers · Email & push · Docker container

▼

Data layer
Supabase · PostgreSQL

~90 relational tables · Row-level security · Managed authentication · Private file storage · Realtime updates · Daily backups

Area	Detail
Frontend	React 19, TypeScript, Vite, Tailwind CSS, TanStack Query, React Router, React Hook Form + Zod, Framer Motion
Backend	Node.js 20+, Express, TypeScript, Zod validation, Pino structured logging, Helmet, CORS, rate limiting
Database & storage	Supabase (PostgreSQL, Auth, Storage, Realtime), versioned SQL migrations
Hosting	Static frontend on a global CDN; backend as a Docker container on any cloud (e.g. Railway, Render, AWS, Google Cloud, Azure)
Quality	Strict TypeScript across both tiers; automated unit tests for core business logic (attendance, fees, exams, homework, leave, tracking, transport and more)

12 IMPLEMENTATION & ROADMAP

From sign-up to go-live

A structured onboarding gets your school running with its own data, users and branding.

1 · Set up
Your school created on the platform, with logo, profile and academic year.

›

2 · Structure
Classes, sections, subjects, timetable and fee plans configured.

›

3 · People
Students and staff imported in bulk; accounts and roles assigned.

›

4 · Transport
Vehicles, drivers, routes and stops mapped; students assigned.

›

5 · Go live
Staff guided by the Quick Tour and certified through Website Knowledge.

On the roadmap

The platform is actively developed. Planned additions include:

Parent–teacher meetings
Online scheduling of PTM slots with reminders.

School gallery
A school-wide photo gallery for events and celebrations.

In-app messaging
Direct, moderated communication between families and teachers.

AI-assisted teaching
Building on today's homework assistant with AI-generated study aids.

Geofenced arrivals
School and home zones for automatic arrival detection.

Support-staff portal
A dedicated workspace for non-teaching staff.

Smart School Management
SCHOOL ERP PLATFORM

See it running at your school.

We would be glad to walk you through a live demonstration tailored to your school — your classes, your routes, your fee structure.

1 · Book a demo
A guided tour of every portal with your team.

2 · Pilot
Start with a few classes or a single bus route.

3 · Roll out
Go school-wide, then add more campuses.

Designed, built and supported by
CraftLanee

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